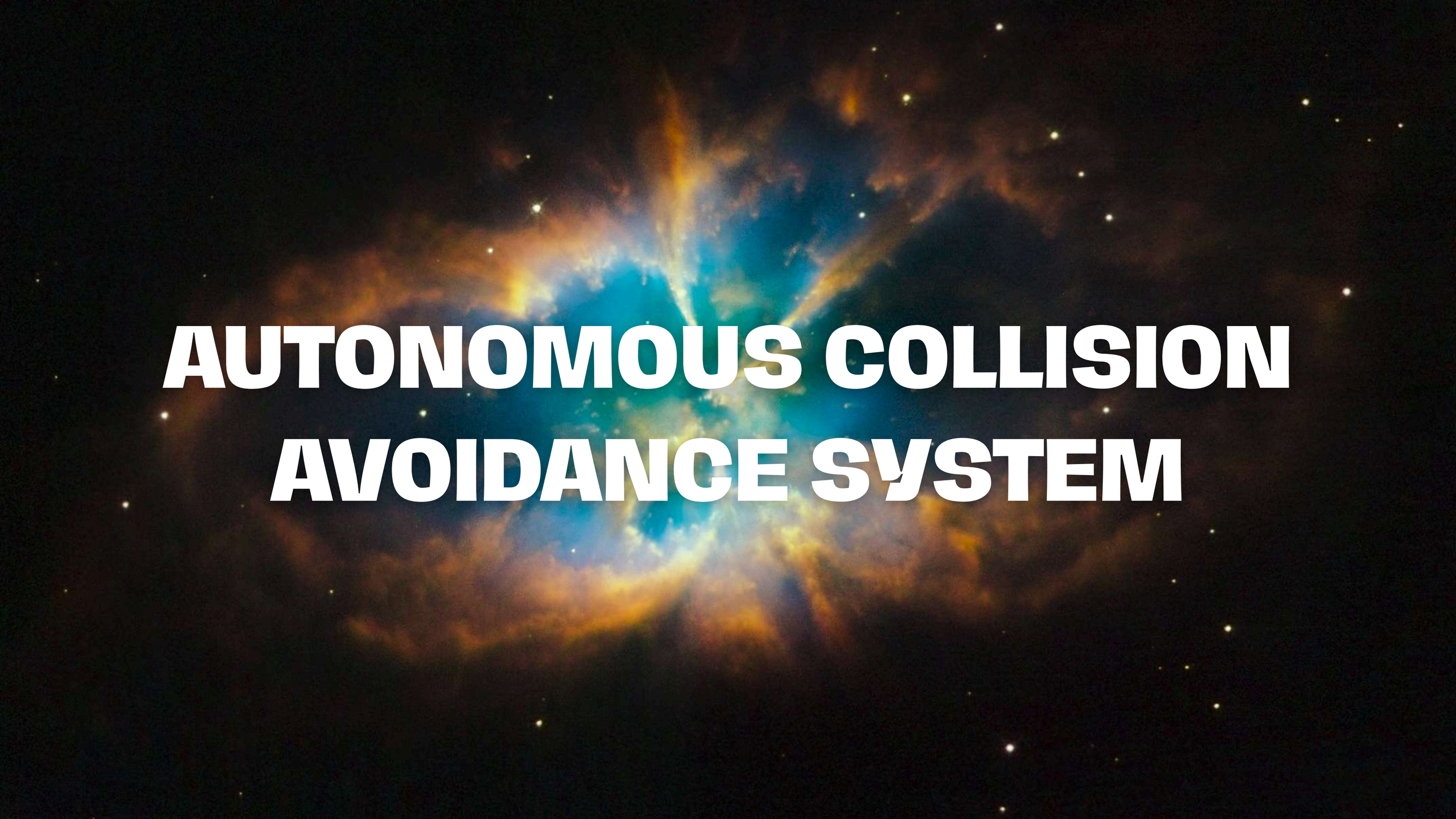
OUR TEAM

WE ARE TEAM OF 4 PEOPLE FROM KILC COLLEGE

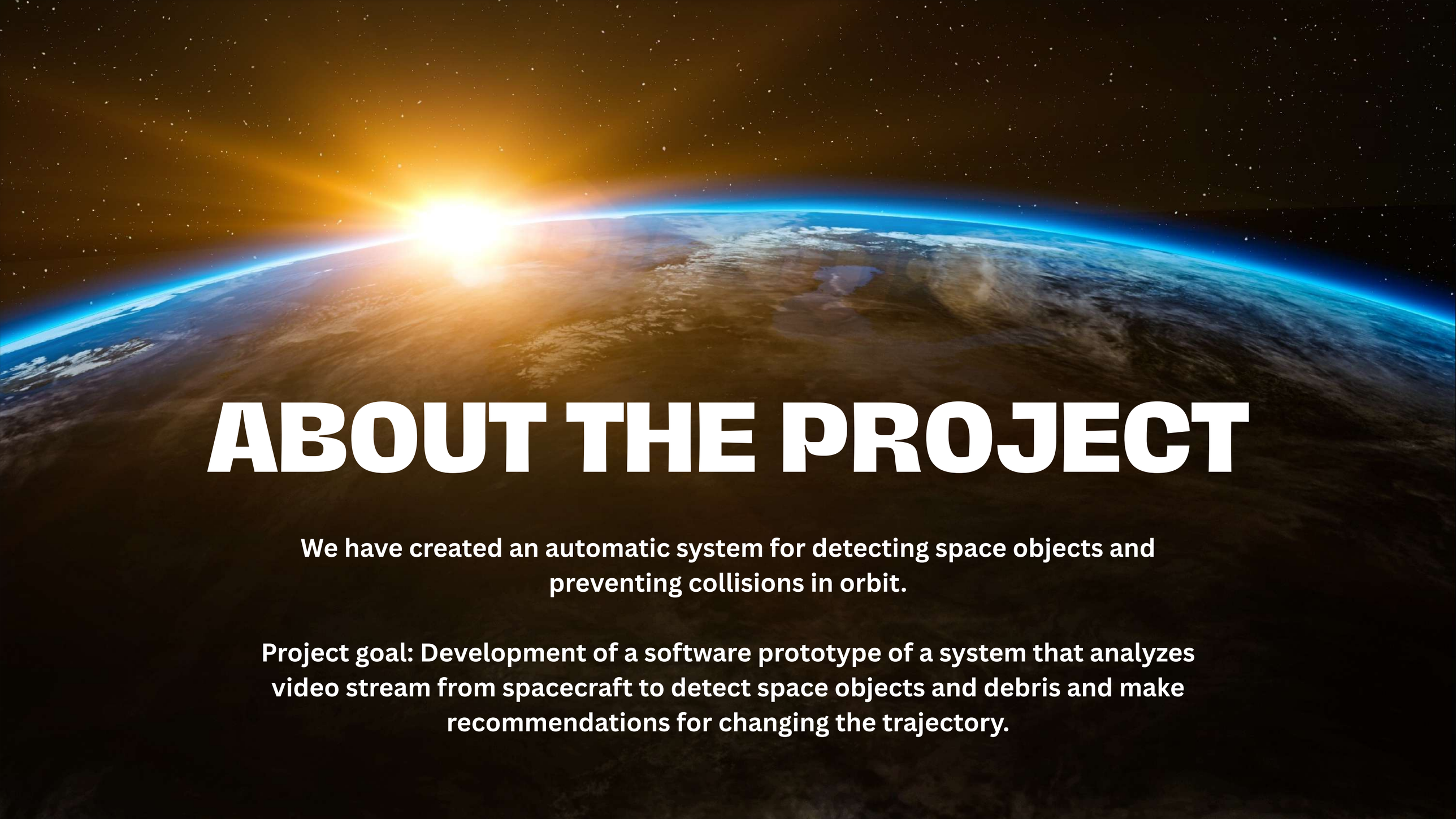
RELEVANCE

Modern cosmonautics faces the problem of man-made objects overload in the low Earth orbit. There are thousands of satellites and millions of fragments of space debris orbiting the Earth. Even a small object can damage or destroy a satellite due to its high speed, making it very dangerous for spacecraft to be in space.

We have a solution for this problem:
We can use computer vision and neural networks to automatically detect objects in outer space.



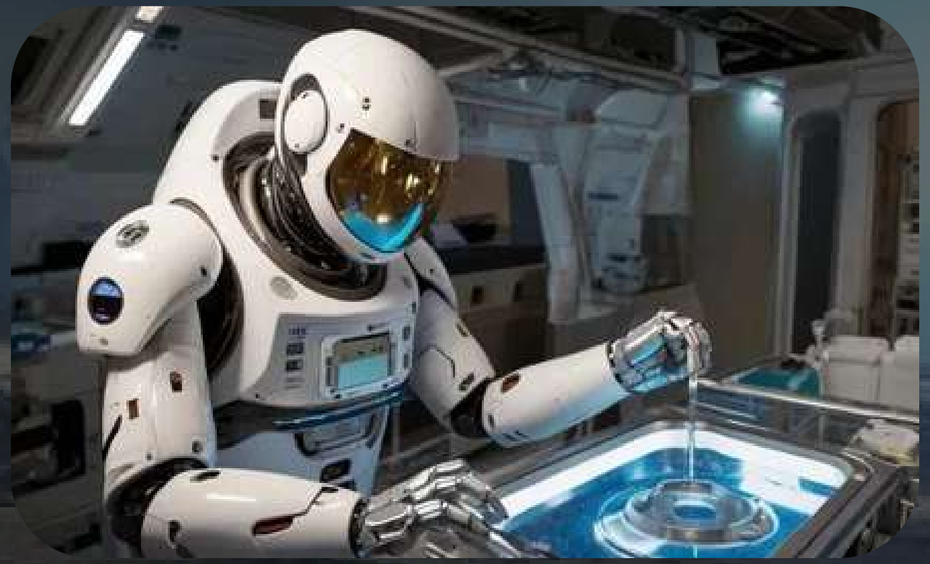
AUTONOMOUS COLLISION AVOIDANCE SYSTEM



ABOUT THE PROJECT

We have created an automatic system for detecting space objects and preventing collisions in orbit.

Project goal: Development of a software prototype of a system that analyzes video stream from spacecraft to detect space objects and debris and make recommendations for changing the trajectory.



GOALS AND TASKS

PROJECT OBJECTIVES:

- Analysis of the space imagery;
- Detection of dangerous space objects;
- Alert direction change to dodge foreign objects.

PROJECT TASKS:

- Use of neural network to detect foreign objects;
- Identification of dangerous types of space objects;
- Allocation of dangerous areas along the course of movement;
- Execution of the decision-making algorithm;
- Visualization is the result of the analysis.

PURPOSE OF THE SYSTEM

The system is designed for use in space tasks, such as:

- Detection of space debris;
- Tracking of other satellites;
- Collision prevention in orbit;
- Increasing the autonomy of spacecraft.

The project is a software prototype and it demonstrates the principles of such systems.

USED PROGRAMS

Our project uses Python, the main language of our program, and also uses technologies and libraries such as:

- YoloV8 - a neural network for object recognition;
- OpenCV - processing of images and video stream;
- NumPy - a complex mathematical computer;
- Pillow - text information display.

These tools allow the system to run in real time.

THREATS

DANGEROUS OBJECTS

Potentially dangerous objects include:

- Other satellites;
- Fragments of space debris;
- Large artificial objects;
- Missile modules and parts;
- Unidentified objects.

All detected objects are analyzed in terms of the risk of collision.

DANGER ZONE

A dangerous zone is formed in the center of the image, corresponding to the intended direction of movement of the satellite.

Objects outside the zone are considered safe;

Objects in the central zone are considered potentially dangerous.

The analysis uses the position of the object relative to the center of the frame, which simulates the direction of orbital movement.

ALGORITHM OF AVOIDING COLLISION

If there are no foreign objects → Course remains

If a foreign object is on the way → Alerts the spacecraft

The system compares the workload of left and right sides

The system chooses the course with the fewest number of objects

RESULTS

- Visualization of the space image;
- Display of detected objects;
- Text recommendations and commands.

These results are made in the operation of the system and all detected objects are analyzed in terms of the risk of collision.

CONCLUSIONS AND PERSPECTIVES

CONCLUSIONS

- We have developed a prototype that detects astronomical objects;
- Computer technologies have been used;
- Crash course prevention system has been realized;
- The use of neutral network in space missions has been approved;

PERSPECTIVES OF DEVELOPMENT

- Measurement of distance and relative velocity of objects;
- Application of orbital mechanics;
- Connection with satellite control systems;
- Use of specialized astronomical sensors.

PROJECT ECONOMICS AND COSTS

Basic expenses

1. Working out
- Development and testing (project team) — 0 tenge
- Training and testing of the AI model - 30,000 tenge (one-time)

2. Infrastructure
- Cloud server — 12,000 tenge / month
- Data storage and database - 3,000 tenge / month
- Domain + hosting — 5,000 tenge / year
- SSL certificate - free of charge

3. Support

- Maintenance and monitoring — 5,000 tenge / month
- AI model update — 8,000 tenge / month

Total expenses

- One-time: ~30,000 tenge
- Monthly: ~28,000–30,000 tenge

MONETIZATION AND ECONOMIC EFFICIENCY

STARTER PACK

$19.9

- OBJECT DETECTION
- RISK ALERTS
- WEB-APPLICATION

PROFESSIONAL PACK

$49.9

- INCREASED ACCURACY
- ANALYSIS OF CRASH COURSE RISKS
- HISTORICAL DATA AND API

CORPORATIVE PACK

$~

- INDIVIDUAL PRICE
- INTEGRATION WITH CLIENT SYSTEMS
- 24/7 SUPPORT

THANKS
FOR ATTENTION